

Geometry, Topology, and Physics – Mikio Nakahara

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1 Quantum Physics

1.1 Analytical Mechanics

1.1.1 Newtonian Mechanics

Equation of motion: $m \frac{d^2 x}{dt^2} = F(x)$

If $F(x) = -\nabla V(x)$, then $V(x)$ is called the **potential** and F is called a **conserved force**. The quantity

$$E = \frac{m}{2} \left(\frac{dx}{dt} \right)^2 + V(x)$$

is conserved: $dE/dt = 0$.

1.1.2 Lagrangian Mechanics

Describe the system with N parameters $\{q_i\}$, the **generalized coordinates**. The generalized velocity is $\dot{q} = dq/dt$.

Let there be a Lagrangian function $L(q, \dot{q})$. Define the action as a functional

$$S[q(t), \dot{q}(t)] = \int_{t_i}^{t_f} L(q, \dot{q}) dt$$

Hamilton's principle: the physical trajectory is that of the least action. Consider a trajectory $q(t)$ of a particle and vary it while holding endpoints fixed; $\delta q(t_i) = \delta q(t_f) = 0$.

$$\begin{aligned} 0 = \delta S &= \int_{t_i}^{t_f} L(q + \delta q, \dot{q} + \delta \dot{q}) dt - \int_{t_i}^{t_f} L(q, \dot{q}) dt \\ &= \int_{t_i}^{t_f} \left(\frac{\partial L}{\partial q} \delta q + \frac{\partial L}{\partial \dot{q}} \frac{d(\delta q)}{dt} \right) dt \\ &= \int_{t_i}^{t_f} \left(\frac{\partial L}{\partial q} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \right) \delta q dt + \underbrace{\left[\frac{\partial L}{\partial \dot{q}} \delta q \right]_{t_i}^{t_f}}_0 \end{aligned}$$

Thus we get the Euler-Lagrange equation

$$\frac{\partial L}{\partial q} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) = 0$$

Define the **generalized momentum** as $p = \partial L / \partial \dot{q}$. Then Euler-Lagrange becomes $\dot{p} = \partial L / \partial q$.

If L is independent of q , then q is called **cyclic**. The momentum conjugate to a cyclic coordinate is conserved:

$$\frac{\partial L}{\partial q} = 0 \longrightarrow \frac{dp}{dt} = 0$$

1.1.3 Hamiltonian Mechanics

Find the Hamiltonian from the Lagrangian using a Legendre transformation:

$$H(q, p) = \sum_k p_k \dot{q}_k - L(q, \dot{q})$$

(q, p) is the **phase space**. From the variation of the Hamiltonian δH , we identify **Hamilton's equations**:

$$\dot{q}_k = \frac{\partial H}{\partial p_k} \quad , \quad \dot{p}_k = -\frac{\partial H}{\partial q_k}$$

Define the Poisson bracket of A, B in phase space as

$$\{A, B\} = \sum_k \left(\frac{\partial A}{\partial q_k} \frac{\partial B}{\partial p_k} - \frac{\partial A}{\partial p_k} \frac{\partial B}{\partial q_k} \right)$$

This is a special case of the **Lie bracket**, and satisfies

1. Linearity: $[A, c_1 B_1 + c_2 B_2] = c_1 [A, B_1] + c_2 [A, B_2]$
2. Skew-symmetry: $[A, B] = -[B, A]$
3. Jacobi identity: $[[A, B], C] + [[B, C], A] + [[C, A], B] = 0$

The fundamental Poisson brackets are

$$\begin{aligned}\{p_i, p_j\} &= \{q_i, q_j\} = 0 \\ \{p_i, q_j\} &= \delta_{ij}\end{aligned}$$

Note that $\{A, H\} = dA/dt$. So if $\{A, H\} = 0$, then A is conserved.

$$\begin{aligned}\{A, H\} &= \sum_k \left(\frac{\partial A}{\partial q_k} \frac{\partial H}{\partial p_k} - \frac{\partial A}{\partial p_k} \frac{\partial H}{\partial q_k} \right) \\ &= \sum_k \left(\frac{\partial A}{\partial q_k} \frac{dq_k}{dt} + \frac{\partial A}{\partial p_k} \frac{dp_k}{dt} \right) = \frac{dA}{dt}\end{aligned}$$

Noether's theorem: Let $H(q_k, p_k)$ be a Hamiltonian invariant under an infinitesimal transformation $q_k \rightarrow q_k + \varepsilon f_k(q)$. Then the following quantity is conserved:

$$Q = \sum_k p_k f_k(q)$$

E.g. for the free-particle harmonic oscillator $H = \frac{1}{2}m\dot{q}^2$, H is obviously invariant under $q \rightarrow q + \varepsilon \cdot 1$, so $Q = p$ is conserved. Indeed, this is linear momentum.

1.2 Quantum Mechanics

Let $\mathcal{H}$ be the Hilbert space (space of wavefunctions/kets normalizable to unity or the Dirac delta function). Represent linear functions $\alpha : \mathcal{H} \rightarrow \mathbb{C}$ as $\langle \alpha |$. The action on kets is $\langle \alpha | \psi \rangle \in \mathbb{C}$. The set of these linear functions is the dual vector space of $\mathcal{H}$, denoted $\mathcal{H}^*$. Elements are called bras, a slightly unfortunate name.

To translate classical equations to quantum (roughly), replace the Poisson bracket with the commutator multiplied by $-i/\hbar$.

$$\begin{aligned}[A, B] &= AB - BA \\ \{A, B\} &\longrightarrow -\frac{i}{\hbar}[A, B]\end{aligned}$$

The canonical relations are

$$\begin{aligned}[X_i, X_j] &= [P_i, P_j] = 0 \\ [X_i, P_j] &= i\hbar\delta_{ij}\end{aligned}$$

1.2.1 The Heisenberg Picture

If $\hat{A}$ is an operator independent of t , from the classical picture, we have Heisenberg's equation of motion:

$$\frac{d\hat{A}}{dt} = -\frac{i}{\hbar}[\hat{A}, H]$$

The solution is $\hat{A}(t) = e^{iHt/\hbar}\hat{A}(0)e^{-iHt/\hbar}$. Let $\hat{U} = e^{-iHt/\hbar}$, so $\hat{A}(t) = \hat{U}^\dagger \hat{A}(0) \hat{U}$. In the Heisenberg picture, operators have time-dependence, but not states.

1.2.2 The Schrodinger picture

States depend on t as $|\psi(t)\rangle = e^{-iHt/\hbar} |\psi(0)\rangle$, but operators don't. By differentiating, we get the Schrodinger equation:

$$i\hbar \frac{d|\psi\rangle}{dt} = \hat{H} |\psi\rangle$$

1.2.3 Position and Momentum Bases

By using the position basis $|x\rangle$, we can expand $|\psi\rangle$ as

$$|\psi\rangle = \int |x\rangle \langle x|\psi\rangle dx = \int \psi(x) |x\rangle dx$$

$\psi(x)$ is called the **wavefunction**. $|\psi(x)|^2$ is the probability to find the particle between x and $x + dx$.

In the x basis (position), we have

$$\hat{x} |x\rangle = x |x\rangle \quad , \quad \hat{p} |x\rangle = -i\hbar \frac{d}{dx} |x\rangle$$

In the p basis (momentum), we have

$$\hat{x} |p\rangle = -i\hbar \frac{d}{dp} |p\rangle \quad , \quad \hat{p} |p\rangle = p |p\rangle$$

From $\langle x|p\rangle = Ce^{ipx/\hbar}$, and by requiring $\langle x|y\rangle = \delta(x - y)$, we have

$$\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar} \quad , \quad \langle p|x\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{-ipx/\hbar}$$

1.2.4 The Time-Independent Schrodinger Equation

Assume the Hamiltonian is of the form $\hat{H} = \hat{p}^2/2m + V(\hat{x})$. The Schrodinger equation becomes

$$i\hbar \frac{d}{dt} \psi(x, t) = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x, t) + V(x) \psi(x, t)$$

If a separable solution $\psi(x, t) = T(t)\phi(x)$ exists, it satisfies

$$T(t) = e^{-iEt/\hbar}$$

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \phi(x) + V(x) \phi(x) = E \phi(x)$$